A. Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science and Engineering		
Programme Name	B.E (Computer Science & Engineering)		
Course Name	Computer Networks	Session	Jan – June 2026
Course Code	24CSE0215	Semester/Batch	4th /2024
L-T-P(Per Week)	3-0-0	Course Credits	03
Pre-requisite	Basic knowledge of Computers.	NHEQF Level	05[1]
Course Coordinator	Dr. Chetna	SDG Number	4,8,9[4]

1. Objectives of the Course

The scope of the course is to provide a foundation for understanding the key aspects of computer network organisation and implementation, and to obtain a theoretical understanding of data communication and computer networks. Students will be introduced to the fundamentals of computer communication, network design, and network operations, providing a strong conceptual foundation aligned with industry-standard networking practices. The objectives of the course are:

- to build an understanding of the fundamental concepts of computer networking.
- to inculcate the skill in students to construct and debug computer networks.
- to develop, implement and manage computer networking systems within an organization.
- to familiarize with current topics such as network management, security and/or other topics.

2. Course Learning Outcomes

After completion of the course, the student should be able to:

	Course Learning Outcome	*POs	NHEQF Level Descriptor[1]	Sessions
CLO01	Define the hardware, software, and components of a network, as well as their interrelations.	PO1, PO2, PO3, PO5	Q1	8
CLO02	Explain the role of reference models and the hierarchical relationship of their respective layers	PO1, PO3, PO4, PO5	Q1, Q2	8
CLO03	Classify the networking protocols and select the appropriate protocol for a particular design.	PO1, PO2, PO3, PO4, PO5, PO7, PO11	Q2	10
CLO04	Examine the concepts and theories of networking with the real-world scenarios.	PO3, PO4, PO5	Q3, Q4	9
CLO05	Design an enterprise network including topologies, protocols, management and security.	PO4, PO5	Q3, Q5, Q6	10
Total Contact Hours				**45**

Revised Bloom's Taxonomy Terminology
* PO's available at (shorturl.at/cryzF)
**Cognitive Level =CL
***Knowledge Categories = KC

CLO-PO Mapping

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	Type of Assessment's[3]
CLO01		H		H		M						Summative
CLO02	H	H	H	M	M	M					H	Summative
CLO03	H	M		H	M	M					M	Summative
CLO04	H	H		H		H						Summative
CLO05	H	H	H		M	M	M				M	Summative

H=High, M=Medium, L=Low

3. Recommended Books:

Textbooks:

B01: Data Communications and Networking' by Forouzan, 5th Edition, 2013.
B02: Computer Networks' by Andrew S. Tanenbaum 5th Edition, Pearson Education,2013.
B03: Data and Computer Communications' by William Stallings, 8th Edition, Pearson,2007.
B04: Security Engineering: A Guide to Building Dependable Distributed Systems, 2nd Edition by Ross J. Anderson

Reference Books:

B05: Computer Networking: A Top-Down Approach', by Kurose and Ross, Pearson Education, 6th Edition,2013.

E-Resources:

- https://library.chitkara.edu.in/subscribed-books.php

4. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	https://nptel.ac.in/courses/106105183
2.	https://nptel.ac.in/courses/106106091
3.	https://nptel.ac.in/courses/106105081
4.	http://www.brainbell.com/tutorials/Networking/
5.	https://learningnetwork.cisco.com/index.jspa?ciscoHome=true
6.	http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-263j-data-communication-networks-fall-2002/lecture-notes/
7.	https://cn-iitr.vlabs.ac.in/

[1] National Higher Education Qualification Framework Level, Refer to annexure
[1] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF
[3]Types of Assessments can be referred from Type of Assessments. Refer to Annexure.
[4]For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

5. Recommended Tools and Platforms

Cisco Packet Tracer-7.3 or above versions

6. Course Plan:

Lecture Number	Topics	Textbook
1	**Detail Discussion of Course Handout (CHO)**	B01-Chpater-1
2-4	**Introduction:** Uses of Computer Networks, Network Hardware Topologies, Collision Domain, Broadcast Domain **Activity (1)** Introduction to Packet Tracer, Introduction of Cables, Network devices: Hub, Switches, Router.	B01-Chapter-1
5-6	**Reference Models:** Seven-Layer OSI architecture, Concepts of Layers, Protocols and Layer interfaces and PDU, TCP/IP Reference Model, Comparison of OSI and TCP/IP reference models	B01-Chapter-2
7-9	**Physical Layer:** Transmission Media (Cable Media), Wireless Media (Cellular Telephone, Satellite Networks) Types of Connecting Devices (Hubs, Switches, Routers) **Activity (2)**	B01-Chapter-7

	To do peer to peer connectivity, assign the IP address and share the resources	
10-11	**Ethernet Frames,** Ethernet and the OSI Model, Ethernet Frame Fields, Ethernet MAC Address, Frame Processing, Multicast MAC Address, Broadcast MAC Address, Unicast MAC Address	B01-Chapter-13 B01-Chapter-13
12-13	**Data Link Layer**: Types of Errors, Redundancy, Error Detection and Correction, CRC, Check Sum, Hamming code & Distance **Activity (3)** Simulation of Network Devices (HUB, Switches, Router) and connect more than two computers using Switch –Star Topology	B01-Chapter-10
14-15	**Multiple Access Protocols**: - Random Access Protocols–ALOHA, CSMA, CSMA/CA, CSMA/CD	B01-Chapter-11 B01-Chapter-12
16-17	**Channelization Protocols:** FDMA, TDMA, CDMA, **Controlled Access Protocols:** Reservation, Polling, Token Passing, Piggybacking	B01-Chapter-11
18-19	**Noisy Channel:** Stop and Wait, Automatic Repeat Request, go-back-n, selective repeat	B01-Chapter-12
ST-1 Syllabus (Lecture number 1-19)		
20-21	**Noiseless Channels:** Elementary data link protocols, Simplex, Stop and Wait Protocol	B01-Chapter-11 B01-Chapter-23
22-25	**Network Layer:** Concept of IP packet and addresses, IPv4 protocol format, Routing Algorithm-Distance Vector Routing, Link State Routing	B01-Chapter-19
26-30	Network Classes (A, B, C, D) and Subnetting, Troubleshooting **Activity (4)** Subnetting of Class A, B and C using different IP addresses, using FLSM and using VLSM	B01-Chapter-21
31-32	ICMP, IGMP, ARP, RARP, IPV6, Transition from IPv4 to IPv6(format) **Activity (5)** Basic commands of Routers: hostname, password, Show Run, Show IP int brief, Assigning IP addresses to interfaces, To Perform Static Routing, Default Routing using 2 and 3 routers	B01-Chapter-20
33-35	**Static and Dynamic routing algorithms**: Shortest Path Routing, Routing Protocols (Static and Dynamic) **Activity (6)** To perform dynamic routing using RIP (Routing Information Protocol) to enable automatic route discovery and exchange of routing information among networked routers, ensuring efficient packet forwarding and adaptability to network topology changes	B01-Chapter-22
36-38	**Routing Protocols:** OSPF, EIGRP, Introduction to BGP, Routing Information Protocol (v1 & v2) **Activity (7)** To perform dynamic routing using OSPF (Open Shortest Path First) to enable routers to automatically discover network topology, calculate the shortest path using link-state information, and efficiently forward packets with fast convergence.	B01-Chapter-22
39-40	**Transport layer:** Services, Connection Less and Connection Oriented protocol, Transport Layer Protocols, TCP Connection, Transmission Control Protocol with Three-Way Handshaking	B01-Chapter-23
41-42	TCP / UDP Message Format, Congestion Control and Quality of Service	B01-Chapter-23
ST-2 Syllabus (Lecture number 1-42)		
43	**Application Layer:** Domain Name System, Remote Logging, Electronic Mail, Introduction to FTP and WWW, Introduction to HTTP and SMTP	B01-Chapter-25
44-45	**Network Security:** Introduction to Cryptography, Public and Private Key, Digital Signature, Internet Protocol Security (IPSec), Firewalls and Types of Firewalls, Pretty Good Protocol (PGP)	B01-Chapter-30,31
ETE (Syllabus = Lecture number 1-45)		

7. **Delivery/Instructional Resources**

Lecture Number	Topics	CLO	Book No, CH No, Page No	TLM[1]	ALM[2]	Web References	Audio-Video
1	**Detailed Discussion of Course Handout (CHO)**	CLO01	B01-Chapter-1	Lecture Discussion			
2-4	**Introduction:** Uses of Computer Networks, Network Hardware Topologies, Collision Domain, Broadcast Domain	CLO01	B01-Chapter-1	Lecture Discussion	Quiz, Challenge, Assignment	https://www.cisco.com/c/en/us/solutions/automation/network-topology.html	https://www.youtube.com/watch?v=uDulBxDb7GM
5-6	**Reference Models:** Seven-Layer OSI architecture, Concepts of Layers, Protocols and Layer interfaces and PDU, TCP/IP Reference Model, Comparison of OSI and TCP/IP reference models	CLO02	B01-Chpater-2	Lecture Discussion	Quiz, Challenge, Assignment	http://www.ics.uci.edu/%7Emagda/Courses/netsys270/ch2_v1.ppt https://slideplayer.com/slide/254123/ https://www.slideshare.net/ankurkumar983/tcp-ip-model	https://www.youtube.com/watch?v=vv4y_uOneC0 https://www.youtube.com/watch?v=2QGgEk20RXM
7-9	**Physical Layer:** Transmission Media (Cable Media), Wireless Media (Cellular Telephone, Satellite Networks) Types of Connecting Devices (Hubs, Switches, Routers)	CLO02	B01-Chapter-7	Lecture Discussion	Quiz, Challenge, Assignment	http://www.ics.uci.edu/%7Emagda/Courses/netsys270/ch1_v1.ppt	https://www.youtube.com/watch?v=BJ7f-HcttyE https://www.youtube.com/watch?v=8ONuDQF7gOY
10-11	**Ethernet Frames,** Ethernet and the OSI Model, Ethernet Frame Fields, Ethernet MAC Address, Frame Processing, Multicast MAC Address, Broadcast MAC Address, Unicast MAC Address	CLO03 CLO04	B01-Chpater-13	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/ethernet-frame-format/ https://www.javatpoint.com/ethernet-frame-format https://help.stonesoft.com/onlinehelp/StoneGate/SMC/6.5.0/GUID-74D81A6F-9EE0-433E-904F-5BC4722B1202.html#:~:text=In%20Ethernet%20(as%20defined%20in,%3Afe%3Afe%20are%20not. https://networklessons.com	https://www.youtube.com/watch?v=ewpq3qxx5Ls https://www.youtube.com/watch?v=TIiQiw7fpsU https://www.youtube.com/watch?v=q2U7Rg9KNG8 https://www.youtube.com/watch?v=GK_uF6cvrOY https://www.youtube.com/watch?v=IUSyV2BVh4A https://www.youtube.com/watch?v=3YcKrtVo2Ro

						/multicast/multicast-ip-address-to-mac-address-mapping	
12-13	**Data Link Layer:** Types of Errors, Redundancy, Error Detection and Correction, CRC, Check Sum, Hamming code & Distance	CLO03 CLO04	B01-Chapter-10	Lecture Discussion	Quiz, Challenge, Assignment	http://www.engppt.com/2009/12/networking-fourozan-ppt-slides.html	https://www.youtube.com/watch?v=eQgRDdBD5Os
14-15	**Multiple Access Protocols**: - Random Access Protocols, ALOHA, CSMA, CSMA/CA, CSMA/CD	CLO03 CLO04	B01-Chpater-11 B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/amogha7/random-access-protocol-in-communication-251294924	https://www.youtube.com/watch?v=YAjfUc7Tt24
16-17	**Channelization Protocols:** FDMA, TDMA, CDMA, Controlled Access Protocols: Reservation, Polling, Token Passing, Piggybacking	CLO03 CLO04	B01-Chpater-11	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/SammarKhan2/fdmatdmacdma	https://www.youtube.com/watch?v=KviHyRss-dE
18-19	**Noisy Channel:** Stop and Wait, Automatic Repeat Request, go-back-n, selective repeat	CLO03 CLO04	B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/Vishal061/unit-2-data-link-control	https://www.youtube.com/watch?v=YdkksvhkQGQ
20-21	**Noiseless Channels:** Elementary data link protocols, Simplex, Stop and Wait Protocol,	CLO03 CLO04	B01-Chpater-11 B01-Chapter-23	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/noiseless-channel-protocol/	https://www.youtube.com/watch?v=n09DfvemnTQ
22-25	**Network Layer:** Concept of IP packet and addresses, **IPv4 protocol format, Routing Algorithm- Distance Vector Routing, Link State Routing**	CLO03 CLO04	B01-Chapter-19	Lecture Discussion	Quiz, Challenge, Assignment	https://www.baeldung.com/cs/ipv4-datagram https://slideplayer.com/slide/4905255/	https://www.youtube.com/watch?v=STJhn9gKF2g
26-30	Network Classes (A, B, C, D) and Subnetting, Troubleshooting	CLO03 CLO04	B01-Chapter-21	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/adkpcte/ip-address https://www.slideshare.net/gichelleamon/subnetting-12046383	https://www.youtube.com/watch?v=0qRcYFGK_60&t=1134s https://www.indiabix.com/networking/subnetting/
31-32	ICMP, IGMP, ARP, RARP, IPV6, Transition from IPv4 to IPv6(format),	CLO03 CLO04	B01-Chapter-20	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/asimnawaz54/internet-control-	https://www.youtube.com/watch?v=xTqtm7-k25o https://www.youtube.com/watch?v=eBHwkyWgVaM

						message-protocol https://www.slideshare.net/satish486/ipv6-17005017 https://www.slideshare.net/raghavendrahamilpure/igmp-35557007	https://www.youtube.com/watch?v=aor29pGhlFE
33-35	**Static and Dynamic routing algorithms**: Shortest Path Routing, **Routing Protocols (Static and Dynamic)**	CLO03 CLO04	B01-Chapter-22	Lecture Discussion	Quiz, Challenge, Assignment	https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_rip/configuration/15-mt/irr-15-mt-book/irr-cfg-info-prot.html	https://www.youtube.com/watch?v=NdjcgVreDDU
36-38	**Routing Protocols:** OSPF, EIGRP, Introduction to BGP, Routing Information Protocol (v1 & v2)	CLO03 CLO04	B01-Chapter-22	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/escrimag/ospfppt-35277878 https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_rip/configuration/15-mt/irr-15-mt-book/irr-cfg-info-prot.html	https://www.youtube.com/watch?v=Zsf9f26rH8U https://www.youtube.com/watch?v=NdjcgVreDDU https://www.youtube.com/watch?v=_Z29ZzKeZHc
39-42	**Transport layer**: Services, Connection Less and Connection Oriented protocol, **Transport Layer Protocols, TCP Connection, Transmission Control Protocol with Three-Way Handshaking, TCP / UDP Message Format, Congestion Control and Quality of Service**	CLO03 CLO04	B01-Chapter-23	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/ahdkhalid/tcp-and-udp https://www.slideshare.net/AlokTripathi40/tcpip-3way-handshake https://www.slideshare.net/tmavroidis/tcpudpicmpandthetransportlayer?qid=d2cf871d-baca-48bf-a3d8-bd1381325b54&v=&b=&from_search=14 https://www.slideshare.net/AmanJaiswal32/congestion-control-68607381	https://www.youtube.com/watch?v=MMDhvHYAF7E https://www.youtube.com/watch?v=LyDqA-dAPW4 https://www.youtube.com/watch?v=uwoD5YsGACg

43	**Application Layer:** Domain Name System, Remote Logging, Electronic Mail, Introduction to FTP and WWW, Introduction to HTTP, SMTP	CLO03 CLO04	B01-Chapter-25	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/siddiqueibrahim37/domain-name-system-29792343 https://www.slideshare.net/BirminghamPublicLIbrary/basic-email https://www.slideshare.net/vinhnguyen509/file-transfer-protocol-36928060 https://www.slideshare.net/hetaljadav/snmp-26639208 https://www.slideshare.net/ToushikPaul/httpprotocol	https://www.youtube.com/watch?v=JkEYOt08-rU https://www.youtube.com/watch?v=pnoWCK82apU https://www.youtube.com/watch?v=GeDhsBRiqro
44-45	**Network Security:** Introduction to Cryptography, Public and Private Key, Digital Signature, Internet Protocol Security (IPSec), Firewalls and Types of Firewalls, Pretty Good Protocol (PGP)	CLO05	B01-Chapter-30,31	Lecture Discussion	Quiz, Challenge, Assignment	https://www.slideshare.net/slideshow/network-security-44529321/44529321	https://youtu.be/5AHp0f0489E

8. Remedial Classes[4]

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

9. Self-Learning[2]

Assignments to promote self-learning, a survey of contents from multiple sources.

S. No.	Topics	CLO	ALM	References/MOOCS
1	Introduction to Computer Networks	CLO01	Group Discussion	https://nptel.ac.in/courses/106105183 https://www.coursera.org/learn/packt-tcp-ip-routing-and-wireless-networking-wvokq
2	Physical Layer & Data Link Layer	CLO02 CLO03	Group Discussion	https://nptel.ac.in/courses/106105183 https://www.coursera.org/learn/packt-tcp-ip-routing-and-wireless-networking-wvokq

[2] Refer to Annexure

3	Network Layer	CLO03 CLO04	Group Discussion	https://nptel.ac.in/courses/106105183 https://www.coursera.org/learn/packt-tcp-ip-routing-and-wireless-networking-wvokq
4	Transport Layer & Application Layer	CLO03 CLO04	Group Discussion	https://nptel.ac.in/courses/106105183 https://www.coursera.org/learn/packt-tcp-ip-routing-and-wireless-networking-wvokq
5	Network Security	CLO05	Group Discussion	https://nptel.ac.in/courses/106105183 https://www.coursera.org/learn/packt-tcp-ip-routing-and-wireless-networking-wvokq

10. Delivery Details of Content Beyond Syllabus[3]

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Everything You Need to Know about Computer Networking and How the Internet Works	CLO01	PO1, PO9	Group Discussion	https://www.taylorfrancis.com/books/mono/10.1 201/9780429447358/internet-book-douglas- comer
2	Computer Networking: Principles, Protocols and Practice	CLO02	PO2	Group Discussion	https://dlib.hust.edu.vn/handle/HUST/22916
3	Using computer networking to facilitate the acquisition of interactive competence	CLO03	PO10	Group Discussion	https://www.sciencedirect.com/science/article/ab s/pii/0346251X9490037X
4	Is the Internet a US invention? An economic and technological history of computer networking	CLO01	PO1, PO9	Group Discussion	https://www.sciencedirect.com/science/article/ab s/pii/S0048733302000690
5	Guide to Computer Network Security	CLO06	PO1, PO4	Group Discussion	https://link.springer.com/book/10.1007/978-3-031-47549-8
6	A Survey on Biologically Inspired Algorithms for Computer Networking	CLO01	PO1, PO9	Group Discussion	https://ieeexplore.ieee.org/abstract/document/64 22291

11. Evaluation Scheme & Components:

Assessment Type[4]	Evaluation Component[5]	Type of Component[6]	No. of Assessments[7]	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Summative	Component 2	Sessional Tests(STs)	02*	50%	50	Computer based online in Campus	1 – 2 1 – 4
Summative	Component 3	End Term Examination	01**	50%	50	Computer based online in Campus	1 - 5
Total				100%	100		

*Students will have to appear in all Sessional Tests.
**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

[3] Refer to Annexure
[4] Refer to Annexure 2 of NCrF
[5] Refer to Annexure
[6] Refer to Annexure
[7] Refer to Annexure

12. Syllabus of the Course:

Subject: Computer Networks / 24CSE0215			
S. No.	**Topic (s)**	**No. of Sessions**	**Weightage %**
1	**Introduction**: Uses of Computer Networks, Network Hardware Topologies, Collision Domain, Broadcast Domain, **Reference Models**: Seven-Layer OSI architecture, Concepts of Layers, Protocols and Layer interfaces and PDU, TCP/IP Reference Model, Comparison of OSI and TCP/IP reference models, **Physical Layer**: Transmission Media (Cable Media), Wireless Media (Cellular Telephone, Satellite Networks) Types of Connecting Devices (Hubs, Switches, Routers), **Ethernet Frames**, Ethernet and the OSI Model, Ethernet Frame Fields, Ethernet MAC Address, Frame Processing, Multicast MAC Address, Broadcast MAC Address , **Data Link Layer**: Types of Errors, Redundancy, Error Detection and Correction, CRC, Check Sum, Hamming code & Distance **Multiple Access Protocols**: - Random Access Protocols-ALOHA, CSMA, CSMA/CA, CSMA/CD **Channelization Protocols**: FDMA, TDMA, CDMA, Controlled Access Protocols: Reservation, Polling, Token Passing, Piggybacking, **Noisy Channel**: Stop and Wait, Automatic Repeat Request, go-back-n, selective repeat	**19**	**42%**
ST-1 (Covering 42% syllabus)			
2	**Noiseless Channels**: Elementary data link protocols, Simplex, Stop and Wait Protocol, **Network Layer**: Concept of IP packet and addresses,IPv4 protocol format, Routing Algorithm-Distance Vector Routing, Link State Routing Network Classes (A, B, C, D) and Subnetting, Troubleshooting, ICMP, IGMP, ARP, RARP, IPV6, Transition from IPv4 to IPv6(format), **Static and Dynamic routing algorithms:** Shortest Path Routing, Routing Protocols (Static and Dynamic), **Routing Protocols:** OSPF, EIGRP, Introduction to BGP, Routing Information Protocol (v1 & v2), **Transport layer:** Services, Connection Less and Connection Oriented protocol, Transport Layer Protocols, TCP Connection, Transmission Control Protocol with Three-Way Handshaking, TCP / UDP Message Format, Congestion Control and Quality of Service	**23**	**93%**
ST-2 (Covering 93% syllabus)			
3	**Application Layer:** Domain Name System, Remote Logging, Electronic Mail, Introduction to FTP and WWW, Introduction to HTTP and SMTP, **Network Security:** Introduction to Cryptography, Public and Private Key, Digital Signature, Internet Protocol Security (IPSec), Firewalls and Types of Firewalls, Pretty Good Protocol (PGP)	**3**	**100%**
End Term 100% syllabus			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Chetna	
Head-Academic Delivery	Dr. Mrinal Paliwal	
Dean	Dr. Rishu Chhabra	
Date	15.01.2026	